Backlyst Credit System Architecture

Ei document ta current codebase onujayi Backlyst er credit system er full architecture, schema, table relationship, business rule, ar current implementation behavior explain kore.

> **Scope:** Ei document e shudhu credit-related schema ar shared table er credit-relevant field gula include kora hoyeche. Shared domain table jemon `User`, `Subscription`, `PlacementTask`, `Website` er full business field na diye credit-relevant part explain kora hoyeche.

1. System Overview

Credit system hocche ekta point wallet + task settlement + audit trail system. User task post korte credit spend kore, provider task complete korle earned credit pete pare, ar sob movement audit hishebe `CreditLog` table e save hoy.

Main Credit Sources

- `Subscription credits`: batch-wise store hoy, expire korte pare
- `Earned task credits`: provider earn kore, requester spend korte pare
- `Visibility credits`: schema te ache, current active flow e kom use hocche

Main Credit Consumers

- Requester task create/post flow
- Task final settlement
- Admin adjustment
- Refund and reversal scenarios

2. Technical Dependencies

- `PrismaService` diye sob credit DB read/write hoy
- `CreditModule` app-level module hisebe register kora
- `AuthService` user create somoy wallet create kore
- `TaskRequesterService` task cost ar reserve amount calculate kore
- `ProviderTaskService` final credit settlement cost decide kore
- `CreditTaskService` actual deduct/refund/reward execute kore
- `CreditService` summary and statement provide kore
- `ReferrerService` referral progress read kore

3. Credit Enums

3.1 CreditType

Value	Meaning
`SUBSCRIPTION_TASK`	Subscription plan theke paoa task credit
`EARNED_TASK`	Task complete kore earn kora credit
`VISIBILITY`	Visibility-based credit bucket

3.2 CreditAction

Value	Meaning
`ADD`	Manual ba system add
`SPEND`	Credit spend
`EXPIRE`	Expired credit
`REFUND`	Refunded credit

Value	Meaning
`REWARD`	Reward hishebe add

3.3 CreditReason

Value	Meaning
`SUBSCRIPTION_PURCHASE`	Subscription purchase related credit
`TASK_COMPLETION`	Task complete hole provider reward
`TASK_POST`	Requester task post/spend/refund flow
`REFERRAL_WEBSITE_VERIFIED`	Referral stage 1 reward
`REFERRAL_FIRST_BACKLINK`	Referral stage 2 reward
`REFERRAL_PAID_SUBSCRIPTION`	Referral stage 3 reward
`MANUAL`	Manual admin/system operation

3.4 ReferralStage

Value	Meaning
`SIGNED_UP`	User signup complete
`WEBSITE_VERIFIED`	Referred website verified
`FIRST_BACKLINK_ACCEPTED`	First backlink accepted
`PAID_SUBSCRIPTION`	Paid subscription activated

4. Core Credit Tables

4.1 UserCreditWallet

Prottek user er base wallet. Subscription credit ekhane direct number hisebe rakha hoy na; earned ar visibility wallet level e thake.

Field	Type	Constraint / Default	Description
`id`	String	PK, `cuid()`	Wallet primary key
`userId`	String	Unique, FK	Wallet owner user id
`earnedTaskCredits`	Int	Default `0`	Earned task credit balance
`visibilityCredits`	Int	Default `0`	Visibility credit balance
`createdAt`	DateTime	Default `now()`	Creation time
`updatedAt`	DateTime	`@updatedAt`	Last update time

4.2 SubscriptionCreditBatch

Subscription credit batch-wise store hoy so that expiry and oldest-first deduction handle kora jay.

Field	Type	Constraint / Default	Description
`id`	String	PK, `cuid()`	Batch primary key
`walletId`	String	FK	Related user wallet
`userId`	String	FK	Owner user id
`subscriptionId`	String?	Nullable FK	Source subscription
`totalCredits`	Int	Required	Original granted amount

Field	Type	Constraint / Default	Description
`remainingCredits`	Int	Required	Current remaining spendable amount
`expiresAt`	DateTime	Required	Expiry time
`createdAt`	DateTime	Default `now()`	Creation time
`updatedAt`	DateTime	`@updatedAt`	Last update time

Indexes: `walletId`, `userId`, `subscriptionId`, `expiresAt`

4.3 CreditLog

Credit system er full ledger table. Sob spend, refund, reward, manual add/reverse ekhane history hishebe store howar kotha.

Field	Type	Constraint / Default	Description
`id`	String	PK, `cuid()`	Log row id
`walletId`	String	FK	Related wallet
`userId`	String	FK	Owner user
`creditType`	Enum	Required	Which bucket changed
`action`	Enum	Required	Spend, refund, reward, etc.
`reason`	Enum	Required	Business reason
`amount`	Int	Required	How much moved
`balanceBefore`	Int	Required	Balance before operation
`balanceAfter`	Int	Required	Balance after operation
`websiteId`	String?	Nullable FK	Related website if any

Field	Type	Constraint / Default	Description
`taskId`	String?	Nullable FK	Related task if any
`referralId`	String?	Nullable FK	Related referral if any
`subscriptionId`	String?	Nullable FK	Related subscription if any
`subscriptionCreditBatchId`	String?	Nullable FK	Related batch if any
`note`	String?	Nullable	Human-readable note
`createdAt`	DateTime	Default `now()`	Log time

Indexes: `(walletId, createdAt)`, `(userId, createdAt)`, `(creditType, createdAt)`, `(reason, createdAt)`, plus foreign-key indexes.

4.4 UserReferral

Referral relationship table. Ek user arek user ke refer korle relation ekhane store hoy.

Field	Type	Constraint / Default	Description
`id`	String	PK, `cuid()`	Referral id
`referrerUserId`	String	FK	User who referred
`referredUserId`	String	Unique, FK	User who was referred
`referredWebsiteId`	String?	Nullable FK	Website tied to referral
`subscriptionId`	String?	Nullable FK	Subscription tied to referral stage
`codeUsed`	String	Required	Referral code used
`currentStage`	Enum	Default `SIGNED_UP`	Current progress stage
`signedUpAt`	DateTime	Default `now()`	Signup timestamp

Field	Type	Constraint / Default	Description
`websiteVerifiedAt`	DateTime?	Nullable	Stage 1 timestamp
`firstBacklinkAt`	DateTime?	Nullable	Stage 2 timestamp
`paidSubscriptionAt`	DateTime?	Nullable	Stage 3 timestamp
`createdAt`	DateTime	Default `now()`	Creation time
`updatedAt`	DateTime	`@updatedAt`	Last update time

4.5 ReferralReward

Referral er specific stage e koto reward dise tar immutable breakdown.

Field	Type	Constraint / Default	Description
`id`	String	PK, `cuid()`	Reward row id
`referralId`	String	FK	Parent referral
`stage`	Enum	Required	Which stage generated reward
`referrerTaskCredits`	Int	Default `0`	Reward to referrer task wallet
`referrerVisibilityCredits`	Int	Default `0`	Reward to referrer visibility wallet
`referredTaskCredits`	Int	Default `0`	Reward to referred user task wallet
`referredVisibilityCredits`	Int	Default `0`	Reward to referred user visibility wallet

Field	Type	Constraint / Default	Description
`createdAt`	DateTime	Default `now()`	Reward issued time

Unique rule: one reward row per `(referralId, stage)`

5. Shared Tables Used by Credit System

5.1 User (credit-relevant fields only)

Field	Type	Description
`id`	String	User primary key, all wallet/subscription/task relation er source
`referralCode`	String?	Referral code for invite flow
`isReferralEnabled`	Boolean	Referral usage on/off flag
`creditWallet`	Relation	One-to-one wallet relation
`creditLogs`	Relation	User ledger rows
`subscriptionCreditBatches`	Relation	User subscription batches
`referralsSent`	Relation	Referrer side records
`referralsReceived`	Relation	Referred side records
`subscriptions`	Relation	User subscriptions

5.2 Subscription

Field	Type	Constraint / Default	Description
`id`	String	PK	Subscription id

Field	Type	Constraint / Default	Description
`userId`	String	FK	Owner user
`planId`	String	FK	Purchased plan
`status`	Enum	Default `PENDING`	ACTIVE, CANCELLED, EXPIRED, PENDING
`startDate`	DateTime	Required	Subscription start
`endDate`	DateTime?	Nullable	Subscription end
`autoRenew`	Boolean	Default `true`	Auto-renew state
`creditBatches`	Relation	-	Credit batches generated by subscription
`creditLogs`	Relation	-	Ledger rows related to subscription

5.3 SubscriptionPlan

Field	Type	Constraint / Default	Description
`id`	String	PK	Plan id
`price`	Decimal	Required	Plan price
`credit`	Int	Required	Plan-level credit amount
`monthlyTaskCreditAllowance`	Int	Default `1`	Monthly grant size hint
`maxTaskCreditCapacity`	Int	Default `5`	Maximum allowed task credit capacity
`billingInterval`	Enum	Default `MONTHLY`	Monthly or yearly plan

Field	Type	Constraint / Default	Description
`isActive`	Boolean	Default `true`	Purchasable or not

5.4 PlacementTask

Task table er credit settlement related fields ekhane thake.

Field	Type	Constraint / Default	Description
`id`	String	PK	Task id
`requesterId`	String	FK	Task requester
`requesterWebsiteId`	String	Required	Task tied website
`selectedProviderId`	String?	Nullable	Assigned provider
`placementType`	Enum	Required	BACKLINK_INSERTION, GUEST_POST, FLEXIBLE
`creditCost`	Int	Required	Initial calculated task cost
`reservedCreditAmount`	Int	Required	Initially reserved amount
`finalCreditCost`	Int?	Nullable	Final settlement cost after delivery
`providerRewardCredits`	Int?	Nullable	How much provider should earn
`lifecycleStatus`	Enum	Default `DRAFT`	Task life cycle state
`submissionStatus`	Enum	Default `NOT_SUBMITTED`	Delivery review state
`creditLogs`	Relation	-	Task-related credit history

5.5 Website (credit-relevant fields only)

Field	Type	Description
`id`	String	Website primary key
`authorId`	String	Website owner, requester relation
`verificationStatus`	Enum	Referral stage and trust flow te use hote pare
`creditLogs`	Relation	Website-linked credit history
`referrals`	Relation	Referral flow te website relation

6. Table Relationships

- One `User` has one `UserCreditWallet`
- One `UserCreditWallet` has many `SubscriptionCreditBatch`
- One `UserCreditWallet` has many `CreditLog`
- One `Subscription` can generate many `SubscriptionCreditBatch`
- One `PlacementTask` can have many `CreditLog` entries
- One `UserReferral` can have many `ReferralReward` rows
- One `UserReferral` can also be referenced by many `CreditLog` rows
- One `Website` can be linked to many `CreditLog` and referral records

7. Business Rules

7.1 Credit Cost Rules

Task Type	Initial Cost	Reserve Rule	Final Settlement
`BACKLINK_INSERTION`	1	1 reserve	Final cost 1
`GUEST_POST`	2	2 reserve	Final cost 2

Task Type	Initial Cost	Reserve Rule	Final Settlement
`FLEXIBLE`	2	2 reserve	Final type backlink hole 1, guest post hole 2

7.2 Spend Order

- Default e age subscription credits use hoy
- Subscription enough na hole earned credits fallback hishebe use hoy
- Caller chaile explicit `SUBSCRIPTION` ba `EARNED` source force korte pare

7.3 Refund Rules

- Original source earned hole refund earned wallet e back jay
- Original source subscription hole active batch e back jay
- Refund operation o ledger e `REFUND` action hishebe record hoy

7.4 Provider Reward Rules

- Provider reward always earned wallet e jay
- Usually provider reward amount = final task cost
- Reward operation `TASK_COMPLETION` reason e log hoy

7.5 Referral Rules

- Referral stage tracking ache: signup, website verified, first backlink accepted, paid subscription
- Stage-wise reward breakdown table ache
- Current codebase e progress read implemented, full reward issuance flow visible na

8. Runtime Flow

8.1 User Registration Flow

1. User create hoy
2. System `UserCreditWallet` create kore with zero balance
3. Tarpor user future spend/reward/refund receive korte pare

8.2 Task Spend Flow

1. Task create time `creditCost` and `reservedCreditAmount` calculate hoy
2. Credit deduction request ashe `CreditTaskService` e
3. Task id valid kina check hoy
4. User wallet load hoy, na thakle create hoy
5. Subscription batch load hoy and oldest expiry first spend hoy
6. Spend log `CreditLog` e save hoy

8.3 Task Delivery Settlement Flow

1. Provider submit korle final task type determine hoy
2. `finalCreditCost` set hoy
3. `providerRewardCredits` set hoy
4. Requester spend and provider reward alada credit operations diye execute hote pare

8.4 Statement Flow

1. User statement request kore
2. Optional filter apply hoy: type, action, reason, website, date range
3. Paginated `CreditLog` rows return hoy

9. Public and Internal Endpoints

Endpoint	Purpose
`GET /credits/me`	User total credit summary
`GET /credits/statement`	Paginated credit statement
`GET /credits/earned`	Earned credit balance
`GET /credits/subscription`	Active subscription credit balance
`GET /credits/referrers/progress`	Referral progress breakdown
`POST /credits/tasks/deduct`	Deduct task credit
`POST /credits/tasks/refund`	Refund task credit
`POST /credits/tasks/reward-provider`	Reward provider earned credit

10. Current Implementation Notes

> **Important:** Current codebase e schema and core credit operations implemented, but kichu orchestration gap ache.

> •Task service cost calculate kore, kintu actual wallet movement separate credit service call diye hoy

- `SubscriptionCreditBatch` read/update visible, but current repo te batch grant/create flow clearly visible na

> •Referral progress read ache, but referral reward issuance er full active flow current repo te fully visible na

11. Plain-English Summary

Short e bolle, Backlyst er credit system hocche user er task point wallet. Requester task korte point spend kore, provider task finish kore point earn kore, subscription thakle batch-wise point use korte pare, ar sob movement ledger e save hoy. Ei structure ta future e referral, expiry, admin adjustment, ar analytics support korar jonno design kora.